

Introduction to machine learning & optimization

Make use of the provided file “bike_data.csv”. Your role is that of an employee of a bike sharing company. The teacher is the CEO of this company. The sales managers want to increase the usage of bikes by giving customers a discount based on a “smart” model. Your tasks are the following:

1. Check and clean the data. Convert the types of variables where you find that suitable, or add variables based on this data.
2. Characterize different types of weather conditions using clustering. Can you think of other (logical) ways to characterize the weather conditions?
3. How are these conditions linked to the number of bike riders? Use the clusters to explain and make use of graphs to support your explanation.
4. The company wants to increase the usage of bikes in off-peak periods. What are the off-peak periods, and how can you define them? Are the off-peak periods (strongly) depending on the weather or not?
5. In order to predict off-peak periods, a model is desired that can only use 3 variables. Make use of principal component analysis to construct this model. What is the performance of this model?
6. You are the one working with the data at this company: which time frame would be the best to plan maintenance in 2013 given this dataset? Assume that the maintenance workers need no frost and 3 working days in a row, and you need to pay them extra if it is misty outside. (Hint: you may want to use some optimization techniques from week 2, but is not absolutely necessary.)

Make a slide-deck of 10 slides maximum which contains only figures and a header per slide. Make a voice-over which explains your findings and advice to the CEO (10 minutes maximum). Upload the video to your portfolio.

Alternatively: color analysis of images, “poster” functionality of Photoshop.